MONODROMY OF SUSTAINED DEFORMATION SPACES WITH ARBITRARILY MANY SLOPES, AND TWO QUESTIONS OF D'ADDEZIO AND VAN HOFTEN

Abstract. Let Y be a completely slope-divisible p-divisible group over a perfect field, with any finite number of slopes. We prove the converse of D'Addezio and van Hoften's monodromy bound on its sustained deformation space: a formally smooth based formal subscheme lies in the strongly Tate-linear space attached to a Frobenius-stable Lie subalgebra $\mathfrak{b}$ if and only if its local monodromy Lie algebra lies in $\mathfrak{b}$, and the strongly Tate-linear space itself has monodromy exactly $\mathfrak{b}$. The result holds at every prime, including $p = 2$, and for every Frobenius-stable $\mathfrak{b}$, normal or not. The strongly Tate-linear space has to be defined using the image of the stabiliser rather than the full finite group of automorphisms. The proof combines three ingredients. The full Baker–Campbell–Hausdorff law is integral after scaling by p^2, and a canonical subnormal chain with abelian quotients reduces the problem to the abelian case. Prismatization identifies torsors under the integral BCH group with isocrystals. Finally, each abelian step is detected by a Kummer extension of p-divisible groups, whose splitting forces the corresponding coordinate to vanish.

This proves D'Addezio–van Hoften's Conjecture 5.5.7 in the formally smooth case and gives an affirmative answer to their Question 8.7.1. For Question 8.7.2, which remains open in general, we reduce it to a global realisation problem for Lie algebras by Hodge tensors, and we settle the case of maximal local monodromy and the ordinary case inside products of GSpin Shimura varieties at odd p, the latter via Jiang's theorem. In particular Question 8.7.2 has an affirmative answer for the ordinary locus of $\mathscr{A}_2$ at every odd prime.

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1. Introduction

Let k be a perfect field of characteristic p and Y a completely slope-divisible p-divisible group over k. D'Addezio and van Hoften [DvH] study the *sustained deformation space* $\mathrm{Def}^{\mathrm{sus}}(Y)$, which is the formal completion of a central leaf at a point with p-divisible group Y. Its strongly Tate-linear formal subschemes are indexed by Frobenius-stable Lie subalgebras $\mathfrak{b}$ of

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the nilpotent algebra

$$\mathfrak{a} = \bigoplus_{i<j} \mathrm{Hom}(V_j, V_i), \qquad V = \mathbb{D}(Y)[1/p] = \bigoplus_i V_i,$$

where V_i is the slope-λ_i part of the isocrystal. They show that if a formal subscheme Z lies in the space attached to $\mathfrak{b}$, then the local monodromy of the universal isocrystal along Z has Lie algebra contained in $\mathfrak{b}$, and they conjecture the converse [DvH, Conj. 5.5.7]. The converse is what is needed to answer their Question 8.7.1, on the propagation of strong Tate-linearity.

In this paper we prove the converse for formally smooth Z, with no restriction on p, on the number of slopes, or on $\mathfrak{b}$. Precise definitions are recalled in Section 2. For now, $D_{\mathfrak{b}} \subseteq \mathrm{Def}^{\mathrm{sus}}(Y)$ denotes the strongly Tate-linear space attached to $\mathfrak{b}$, and $G(M_Z, z)$ denotes the Tannakian monodromy group, at the base point z, of the rational covariant Dieudonné isocrystal M_Z on Z.

Theorem A (Theorem 2.3)**.** *Let $\mathfrak{b} \subseteq \mathfrak{a}$ be a Frobenius-stable Lie K_0-subalgebra, and let $Z = \mathrm{Spf}\, R \hookrightarrow \mathrm{Def}^{\mathrm{sus}}(Y)$ be a closed formal subscheme through the base point with $R \simeq k[[t_1, \ldots, t_d]]$. Then*

$$Z \subseteq D_{\mathfrak{b}} \iff \mathrm{Lie}\, G(M_Z, z) \subseteq \mathfrak{b},$$

as closed formal subschemes of $\mathrm{Def}^{\mathrm{sus}}(Y)$. Moreover $\mathrm{Lie}\, G(M_{D_{\mathfrak{b}}}, z) = \mathfrak{b}$.

Two points about the statement deserve comment. First, $D_{\mathfrak{b}}$ is the quotient of the naive homogeneous space $Z(p^2\mathfrak{b}^+)$ by the image $H_{\mathfrak{b},2}$ of the stabiliser of $\mathfrak{b}$ in $\mathrm{Aut}(Y)^\circ$, not by the full finite group H which appears in $\mathrm{Def}^{\mathrm{sus}}(Y) = A\backslash T_{\mathfrak{a}}$. Only the elements of A that preserve $T_{\mathfrak{b}}$ act on it, so the natural object is $A_{\mathfrak{b}}\backslash T_{\mathfrak{b}}$ with $A_{\mathfrak{b}} = A \cap T_{\mathfrak{b}}$, and this is what we use. Second, when there are three or more slopes the algebra $\mathfrak{a}$ is not of class two, and $\mathfrak{b}$ need be neither abelian nor normal. We therefore work throughout with the full Baker–Campbell–Hausdorff law. Example 3.6 shows that the higher terms do occur, already at $p = 2$ and $p = 3$.

Applications to the questions of D'Addezio and van Hoften. Let $C = C(Y, \lambda)$ be a central leaf in a Siegel modular variety $\mathscr{A}_g$ over $\overline{\mathbf{F}}_p$. A smooth irreducible closed subvariety $Z \subseteq C$ is *strongly Tate-linear* at a closed point z if its completion $\widehat{Z}_z$ is a strongly Tate-linear formal subscheme of $\widehat{C}_z \simeq \mathrm{Def}^{\mathrm{sus}}(Y_z)$. In [DvH, §8.7] the authors ask:

> **Question 8.7.1.** If Z is strongly Tate-linear at some closed point, is it strongly Tate-linear at every closed point?
>
> **Question 8.7.2.** If Z is strongly Tate-linear at some closed point, is Z an irreducible component of a central leaf in the mod p reduction of a Shimura variety of Hodge type?

They observe that Question 8.7.1 would follow from Conjecture 5.5.7 together with their local monodromy theorem. Our theorem covers exactly the smooth subvarieties which occur there, so we obtain:

Theorem B (Theorem 8.2)**.** *Question 8.7.1 has an affirmative answer.*

Question 8.7.2 is of a different nature. Once Theorems A and B are known, the completed local ring of Z is determined everywhere by one Frobenius-stable Lie algebra $\mathfrak{b}$. The question then becomes whether $\mathfrak{b}$ comes from global Hodge tensors (Proposition 8.3). We do not settle this in general. We do settle the following cases.

Theorem C (Theorems 8.4 and 8.5 and Corollary 8.6)**.** *Let $Z \subseteq C$ be smooth, irreducible, closed, and strongly Tate-linear at one closed point. Question 8.7.2 has an affirmative answer for Z in each of the following cases:*

(i) *the local monodromy of Z is maximal, that is,* $\operatorname{Lie} G(M_Z, z) = \mathfrak{a}$*;*
(ii) *p is odd, C is the ordinary locus, and Z lies in the ordinary locus of the reduction of a product of* GSpin *Shimura varieties.*

In particular, for p odd, every smooth irreducible closed subvariety of $\mathscr{A}_2^{\mathrm{ord}}$ which is strongly Tate-linear at one point is an irreducible component of a central leaf in the reduction of a Hodge-type Shimura variety.

Case (ii) rests on Jiang's proof of Chai's Tate-linear conjecture for products of GSpin Shimura varieties [Jia23]. We also record a liftable ordinary case which follows from Moonen's linearity theorem [Moo98] (Proposition 8.7).

Outline of the proof of Theorem A. In Section 2 we construct $D_{\mathfrak{l}}$ for every Frobenius-stable $\mathfrak{l}$ and show that it is a formal power series scheme, finite flat under the naive homogeneous space $D_{\mathfrak{l},2}$ and closed in $\mathrm{Def}^{\mathrm{sus}}(Y)$.

The upper bound, $Z \subseteq D_{\mathfrak{b}} \Rightarrow \operatorname{Lie} G(M_Z, z) \subseteq \mathfrak{b}$, is proved in Section 5. On $D_{\mathfrak{b},2}$ the universal framing is a torsor under the integral group $\Pi(p^2\mathfrak{b}^+)$. Its period coordinate defines a homomorphism to the smooth integral group U_L obtained from the full BCH law on $L = p^2\mathfrak{b}^+$ (Section 3). After prismatization the torsor therefore induces a tensor functor from representations of $U(\mathfrak{b}) = \exp(\mathfrak{b})$ to isocrystals, and Tannaka duality gives the bound. The comparison between this tensor functor and the actual Dieudonné isocrystal is the only place where prismatic input is used (Section 4).

For the converse we first reduce to abelian steps. If $\gamma_i(\mathfrak{a})$ is the lower central series, then $\mathfrak{h}_i = \mathfrak{b} + \gamma_i(\mathfrak{a})$ is a chain from $\mathfrak{a}$ down to $\mathfrak{b}$ in which each term is normal in the previous one with abelian quotient (Lemma 3.4). So it suffices to treat $\mathfrak{j} \lhd \mathfrak{l}$ with $\mathfrak{d} = \mathfrak{l}/\mathfrak{j}$ abelian. In Section 6, the quotient lattice defines a connected p-divisible group Q_0 and a quotient coordinate $q_0 \colon Z(p^2\mathfrak{l}^+) \to Q_0$ whose zero fibre is $Z(p^2\mathfrak{j}^+)$. The coordinate defines a Kummer extension $0 \to Q_0 \to E_q \to \mathbf{Q}_p/\mathbf{Z}_p \to 0$, and we show that its isocrystal is the image of an explicit representation of $U(\mathfrak{l})$ with kernel $U(\mathfrak{j})$. If the monodromy lies in $\mathfrak{j}$, the Kummer isocrystal therefore splits. Full faithfulness of the Dieudonné functor turns this splitting into a quasi-homomorphism with a single denominator p^N. From it we get compatible p^m-th roots of $[p^N]q$ for all m, and this forces $q = 0$ because $\bigcap_m R^{p^m} = k$. Some care is needed to keep everything on power series bases. We lift through the finite connected quotient by a relative Frobenius instead of a fibre product, and then descend the vanishing of ideals along that Frobenius.

Iterating along the chain gives the converse (Section 7). Exact monodromy follows by applying the converse to $D_{\mathfrak{b}}$ itself, together with a dimension count: for a negative-slope lattice P, $\operatorname{length}_W(P/\varphi^{-1}P) > 0$ unless $P = 0$.

References for the inputs. We use D'Addezio–van Hoften [DvH] for the definition of sustained deformation spaces, the Dieudonné comparison on quasi-regular semiperfect algebras, the upper bound and the local monodromy theorem. Prismatic inputs are from [BS22, BL22], and crystalline Dieudonné theory from [dJ95, Zin02].

2. Setup and the strongly Tate-linear spaces

2.1. **Conventions.** Let k be a perfect field of characteristic p, $W = W(k)$ and $K_0 = W[1/p]$. Let Y be a completely slope-divisible p-divisible group over k with distinct slopes $\lambda_1 > \lambda_2 > \cdots > \lambda_r$. Put

$$V^+ = \mathbb{D}(Y), \qquad V = V^+[1/p] = \bigoplus_{i=1}^{r} V_i.$$

We use the normalised covariant Dieudonné module

$$\mathbb{D}(T) = M_{\mathbb{\Delta}}(T^\vee), \qquad \varphi_{\mathbb{D}} = \varphi_{\mathbb{\Delta}}/p.$$

An integral Dieudonné lattice L thus satisfies $L \subseteq \varphi L \subseteq p^{-1}L$, and the operators which are integral on such lattices are

$$\varphi^{-1} \quad \text{and} \quad p\varphi. \tag{2.1}$$

The normalised φ itself may have denominators, and this will matter in Section 4.

Let

$$\mathfrak{a}_{ij} = \mathrm{Hom}(V_j, V_i), \qquad \mathfrak{a} = \bigoplus_{i<j} \mathfrak{a}_{ij}. \tag{2.2}$$

This is the algebra of strictly slope-lowering endomorphisms of V, and as an associative algebra

$$\mathfrak{a}^r = 0. \tag{2.3}$$

Let $\mathfrak{a}^+ = \mathbb{D}(\mathcal{H}^\circ_Y)$ be the integral lattice of the internal Hom p-divisible group of negative slope.

Notation 2.1. For a completely slope-divisible Dieudonné lattice M of negative slopes, $X(M)$ denotes the connected p-divisible group with $\mathbb{D}(X(M)) = M$, and $\widetilde{X}(M)$ its universal cover. For an F-stable Lie subalgebra $\mathfrak{l} \subseteq \mathfrak{a}$ put

$$\mathfrak{l}^+ = \mathfrak{l} \cap \mathfrak{a}^+, \qquad L_{\mathfrak{l}} = p^2\mathfrak{l}^+, \qquad N_{\mathfrak{l}} = \Pi(L_{\mathfrak{l}}), \qquad T_{\mathfrak{l}} = \widetilde{\Pi}(\mathfrak{l}). \tag{2.4}$$

Here $\Pi(L)$ is the Tate module $T_pX(L)$ and $\widetilde{\Pi}(\mathfrak{l})$ is $\widetilde{X}(\mathfrak{l}^+)$, each with the group law given by the Baker–Campbell–Hausdorff formula (Section 3). On rational points $\widetilde{\Pi}(\mathfrak{l})$ is the unipotent group $U(\mathfrak{l}) = \exp(\mathfrak{l})$. For a lattice L as above, $Z(L) = \Pi(L)\backslash\widetilde{\Pi}(L[1/p])$ is the corresponding homogeneous space.

The sustained deformation space is the fpqc quotient

$$\mathrm{Def}^{\mathrm{sus}}(Y) = A\backslash T_{\mathfrak{a}}, \qquad A = \mathrm{Aut}(Y)^\circ. \tag{2.5}$$

The congruence subgroup $N_{\mathfrak{a}}$ is the kernel of reduction to $Y[p^2]$, and there is a finite group scheme H with

$$1 \to N_{\mathfrak{a}} \to A \to H \to 1. \tag{2.6}$$

For general $\mathfrak{l}$ put $A_{\mathfrak{l}} = A \cap T_{\mathfrak{l}}$ and let $H_{\mathfrak{l},2}$ be the scheme-theoretic image of $A_{\mathfrak{l}} \to H$.

Definition 2.2. The *strongly Tate-linear space* attached to $\mathfrak{l}$ is

$$D_{\mathfrak{l}} = A_{\mathfrak{l}}\backslash T_{\mathfrak{l}} = Z(p^2\mathfrak{l}^+)/H_{\mathfrak{l},2}.$$

For a based formal scheme Z over $\mathrm{Def}^{\mathrm{sus}}(Y)$ we write M_Z for the pullback of the algebraised universal rational covariant Dieudonné isocrystal, identify its fibre at the base point z with V, and write $G(M_Z, z) \subseteq \mathrm{GL}(V)$ for its Tannakian monodromy group.

2.2. **Main theorem.**

Theorem 2.3. *Let $\mathfrak{b} \subseteq \mathfrak{a}$ be any F-stable Lie K_0-subalgebra. Let $Z = \mathrm{Spf}\, R \hookrightarrow \mathrm{Def}^{\mathrm{sus}}(Y)$ be a based formally smooth closed formal subscheme, with R a complete regular local k-algebra with residue field k, so that $R \simeq k[[t_1, \dots, t_d]]$. Then, as closed formal subschemes of* $\mathrm{Def}^{\mathrm{sus}}(Y)$,

$$Z \subseteq D_{\mathfrak{b}} \iff \mathrm{Lie}\, G(M_Z, z) \subseteq \mathfrak{b}. \tag{2.7}$$

Moreover

$$\mathrm{Lie}\, G(M_{D_{\mathfrak{b}}}, z) = \mathfrak{b}. \tag{2.8}$$

There is no hypothesis $p > r$, no bound on the nilpotency class and no condition on how $\mathfrak{b}$ sits with respect to the slope decomposition.

Remark 2.4. As stated, [DvH, Conj. 5.5.7] allows an arbitrary closed immersion $Z \hookrightarrow \mathrm{Def}^{\mathrm{sus}}(Y)$. We only treat formally smooth Z. This is enough for Question 8.7.1, which starts with a smooth subvariety, all of whose completed local rings are power series rings.

2.3. **The quotient by the stabiliser.** For $L_{\mathfrak{l}} = p^2\mathfrak{l}^+$ the naive homogeneous space is

$$D_{\mathfrak{l},2} = N_{\mathfrak{l}}\backslash T_{\mathfrak{l}}. \tag{2.9}$$

After reduction modulo p^2, the stabiliser of $T_{\mathfrak{l}}$ in A is not all of H in general, but only

$$H_{\mathfrak{l},2} = \mathrm{im}(A_{\mathfrak{l}} \to H). \tag{2.10}$$

We will see in the proof of Proposition 2.5 that there is an exact sequence of fpqc sheaves

$$1 \to N_{\mathfrak{l}} \to A_{\mathfrak{l}} \to H_{\mathfrak{l},2} \to 1. \tag{2.11}$$

Consequently $H_{\mathfrak{l},2}$ acts freely on $D_{\mathfrak{l},2}$, and

$$D_{\mathfrak{l}} = D_{\mathfrak{l},2}/H_{\mathfrak{l},2} = A_{\mathfrak{l}}\backslash T_{\mathfrak{l}}. \tag{2.12}$$

This is a quotient of homogeneous spaces. We do not need $A_{\mathfrak{l}}$ to be normal in $T_{\mathfrak{l}}$.

Proposition 2.5. *For every F-stable $\mathfrak{l} \subseteq \mathfrak{a}$, $D_{\mathfrak{l}}$ is a finite-dimensional formally smooth formal power series scheme over k. The map*

$$\pi_{\mathfrak{l}} \colon D_{\mathfrak{l},2} \to D_{\mathfrak{l}} \tag{2.13}$$

is finite and faithfully flat, and the natural map

$$D_{\mathfrak{l}} \hookrightarrow \mathrm{Def}^{\mathrm{sus}}(Y) \tag{2.14}$$

is a closed immersion. Moreover

$$D_{\mathfrak{l}}^{\mathrm{perf}} \simeq D_{\mathfrak{l},2}^{\mathrm{perf}} \simeq T_{\mathfrak{l}}. \tag{2.15}$$

Proof. The lattice $\mathfrak{l}^+ = \mathfrak{l} \cap \mathfrak{a}^+$ is saturated in $\mathfrak{a}^+$. Hence, after scaling by p^2, the kernel of the map from the ambient Tate module to the quotient Tate module is $T_pX(p^2\mathfrak{l}^+)$, and on fpqc test rings

$$T_pX(p^2\mathfrak{a}^+) \cap \widetilde{X}(\mathfrak{l}) = T_pX(p^2\mathfrak{l}^+). \tag{2.16}$$

Combined with the identification of the congruence subgroup in Lemma 3.3, this gives (2.11).

The finite group $H_{\mathfrak{l},2}$ is connected. It is the scheme-theoretic image of a subgroup of $\mathrm{Aut}(Y)^\circ$, and the finite-level images of $\mathrm{Aut}(Y)^\circ$ have a single geometric point, so the augmentation ideal of $\mathcal{O}(H_{\mathfrak{l},2})$ is nilpotent. Write

$$D_{\mathfrak{l},2} = \mathrm{Spf}\, B_2, \qquad B_2 = k[[x_1, \dots, x_e]].$$

Since $\mathcal{O}(H_{\mathfrak{l},2})$ is finite-dimensional, the formal action is induced by an algebraic coaction $B_2 \to B_2 \otimes_k \mathcal{O}(H_{\mathfrak{l},2})$. The action on $\mathrm{Spec}\, B_2$ is free: the stabiliser is finite over B_2 and its augmentation module vanishes at the closed point, hence everywhere by Nakayama. The affine fppf quotient

$$\mathrm{Spec}\, B_2 \to \mathrm{Spec}\, B, \qquad B = B_2^{H_{\mathfrak{l},2}}, \tag{2.17}$$

is therefore finite and faithfully flat. Choose s such that every element of the augmentation ideal of $\mathcal{O}(H_{\mathfrak{l},2})$ has vanishing p^s-th power. Then

$$B_2^{p^s} \subseteq B \subseteq B_2. \tag{2.18}$$

As k is perfect, $B_2^{p^s} = k[[x_1^{p^s}, \dots, x_e^{p^s}]]$, and B_2 is finite over it. So B is a complete Noetherian local ring with residue field k. Regularity descends along the faithfully flat map $B \to B_2$ [Stacks, Tag 07NG], so $B \simeq k[[z_1, \dots, z_e]]$. Completing the algebraic torsor along an invariant primary ideal recovers the formal action, which proves (2.12) and (2.13).

Next, the map to $\mathrm{Def}^{\mathrm{sus}}(Y)$ is a monomorphism. If $u, v \in T_{\mathfrak{l}}$ have the same image, then fpqc-locally $v = au$ for a unique $a \in A$, and $a = vu^{-1} \in T_{\mathfrak{l}}$, so $a \in A_{\mathfrak{l}}$. A based monomorphism of complete Noetherian local formal schemes is a closed immersion: injectivity on dual-number points gives surjectivity on cotangent spaces, so $\mathfrak{m}_A B = \mathfrak{m}_B$ by Nakayama, and successive approximation shows that the map of complete local rings is surjective.

Finally, (2.18) gives equality after perfection, and $D_{\mathfrak{l},2}^{\mathrm{perf}} \simeq T_{\mathfrak{l}}$ is the corresponding statement for the naive homogeneous space. This proves (2.15). □

3. Nilpotent Lie theory

This section collects the facts about nilpotent Lie algebras that are needed when $r \geq 3$. Nothing in it uses prismatic input.

3.1. **The full BCH law.** Because $\mathfrak{a}^r = 0$, the matrix exponential and logarithm are finite sums,

$$\exp(x) = \sum_{m=0}^{r-1} \frac{x^m}{m!}, \qquad \log(1+x) = \sum_{m=1}^{r-1} \frac{(-1)^{m+1} x^m}{m}, \tag{3.1}$$

and $\mathrm{BCH}(x, y) = \log(\exp(x)\exp(y))$. All nested commutators of length $< r$ have to be kept.

Lemma 3.1. *Let M be a nilpotent Dieudonné–Lie lattice and $L = p^2 M$. At every prime p, each Dynkin summand of the BCH polynomial maps $L \times L$ to L, independently of the nilpotency class.*

Proof. A Dynkin term of degree d has, up to sign, denominator

$$D = d\, n \prod_{\nu=1}^{n} a_\nu!\, b_\nu!, \qquad \sum_{\nu} (a_\nu + b_\nu) = d, \quad a_\nu + b_\nu > 0. \tag{3.2}$$

Let t be the number of nonzero entries among the a_ν, b_ν, so that $d \geq t \geq n \geq 1$. Since $v_p(m!) \leq m - 1$ and $v_p(m) \leq m - 1$ for every $m \geq 1$,

$$v_p(D) \leq (d-1) + (n-1) + (d-t) \leq 2d - 2. \tag{3.3}$$

On the other hand $[L, L] \subseteq p^2 L$, so a bracket word with d inputs from L lies in $p^{2d-2} L$. Thus each coefficient is cancelled separately, and every Dynkin summand is an integral polynomial in coordinates on L, even over rings with p-torsion. Terms of length $\geq r$ vanish, so the law is a finite sum. □

Remark 3.2. The series (3.1) must be truncated using the associative nilpotence $\mathfrak{a}^r = 0$, not the Lie nilpotency class of the subalgebra in question. An abelian Lie subalgebra can contain a matrix X with $X^m \neq 0$ for large m.

3.2. **Tate automorphisms and the congruence subgroup.** Lemma 3.1 concerns the abstract BCH lattice. In the ring of Tate endomorphisms one has, for f strictly slope-lowering and $m \geq 1$,

$$v_p\Big(\frac{p^{2m}}{m!}\Big) \geq m + 1 \geq 2, \qquad v_p\Big(\frac{p^{2m}}{m}\Big) \geq m + 1 \geq 2. \tag{3.4}$$

Lemma 3.3. *The exponential induces an isomorphism*

$$\exp\colon \Pi(p^2\mathfrak{a}^+) \xrightarrow{\sim} \ker\bigl(\mathrm{Aut}(Y)^\circ \to \mathrm{Aut}(Y[p^2])\bigr). \tag{3.5}$$

Proof. By (3.4), $\exp(p^2 f) - 1$ and $\log(1 + p^2 f)$ lie in $p^2 T_p \mathcal{H}^\circ_Y$. A homomorphism vanishing on $Y[p^2]$ factors uniquely through $[p^2]\colon Y \to Y$, and the quotient is again slope-lowering. □

This is where the finite group H of (2.6) comes from. Note that (3.5) is a statement about Tate endomorphisms of Y. It does not give an integral embedding $\mathfrak{a}^+ \hookrightarrow \mathrm{End}_W(\mathbb{D}(Y))$, and in arbitrary nilpotency class one should not expect one. We never use such an embedding.

3.3. **A subnormal chain.** An arbitrary $\mathfrak{b} \subseteq \mathfrak{a}$ need not be normal. Let $\gamma_1(\mathfrak{a}) = \mathfrak{a}$ and $\gamma_{i+1}(\mathfrak{a}) = [\mathfrak{a}, \gamma_i(\mathfrak{a})]$ be the lower central series, and put

$$\mathfrak{h}_i = \mathfrak{b} + \gamma_i(\mathfrak{a}), \qquad 1 \le i \le r. \tag{3.6}$$

Lemma 3.4. *One has $\mathfrak{h}_1 = \mathfrak{a}$ and $\mathfrak{h}_r = \mathfrak{b}$, and for $i < r$*

$$\mathfrak{h}_{i+1} \lhd \mathfrak{h}_i, \qquad [\mathfrak{h}_i, \mathfrak{h}_i] \subseteq \mathfrak{h}_{i+1}. \tag{3.7}$$

Every $\mathfrak{h}_i$ is F-stable. The lattices $\mathfrak{h}_i^+ = \mathfrak{h}_i \cap \mathfrak{a}^+$ are saturated and completely slope-divisible, and the quotient lattice

$$D_i = p^2(\mathfrak{h}_i^+ / \mathfrak{h}_{i+1}^+) \tag{3.8}$$

is finite free, completely slope-divisible, of strictly negative slopes, and has zero bracket.

Proof. By the Jacobi identity each $\gamma_i(\mathfrak{a})$ is an ideal. A Lie word of length i is a sum of associative words of length i, so $\gamma_i(\mathfrak{a}) \subseteq \mathfrak{a}^i$ and $\gamma_r(\mathfrak{a}) = 0$. Moreover

$$[\mathfrak{b} + \gamma_i, \mathfrak{b} + \gamma_i] \subseteq [\mathfrak{b}, \mathfrak{b}] + [\mathfrak{a}, \gamma_i] \subseteq \mathfrak{b} + \gamma_{i+1},$$

which gives both normality and commutativity of the quotients. Frobenius preserves the construction.

Saturation follows from the fact that $\mathfrak{h}_i^+$ is the intersection of $\mathfrak{a}^+$ with a rational subspace: if $px \in \mathfrak{h}_{i+1}^+$ and $x \in \mathfrak{h}_i^+$, then $x \in \mathfrak{h}_{i+1} \cap \mathfrak{a}^+$. The integral operators φ^{-1} and $p\varphi$ preserve these sublattices and pass to the quotient. On each slope part, complete slope divisibility restricts to the intersections and passes to the saturated quotient. The slopes remain strictly negative, and the bracket on the quotient vanishes by (3.7). □

3.4. **The integral BCH group.**

Lemma 3.5. *For $L = p^2\mathfrak{l}^+$, affine space on the W-module L with the full BCH law is a smooth affine group scheme U_L over W, with generic fibre $U(\mathfrak{l}) = \exp(\mathfrak{l})$. Every finite-dimensional algebraic representation of $U(\mathfrak{l})$ has a U_L-stable finite free W-lattice.*

Proof. By Lemma 3.1 the multiplication is an integral polynomial law. The identity is 0 and the inverse is $x \mapsto -x$. Associativity holds rationally, hence integrally since the coordinate ring is p-torsion-free. The underlying scheme is affine space, so it is smooth.

For the lattice, write $\mathcal{O}(U_L) = W[X_1, \dots, X_e]$, choose a basis $v_1, \dots, v_a$ of a representation T over K_0, and write the coaction as a finite sum $\delta(v_i) = \sum_\alpha v_{i,\alpha} \otimes X^\alpha$. Put

$$\Lambda = \sum_{i,\alpha} W\, v_{i,\alpha}. \tag{3.9}$$

The counit axiom shows that Λ spans T. Comparing coefficients in the last factor of the coassociativity identity shows that $\delta(\Lambda) \subseteq \Lambda \otimes_W \mathcal{O}(U_L)$, because the BCH comultiplication has integral coefficients. As W is a discrete valuation ring, the finitely generated torsion-free module Λ is free. □

The stable lattice of Lemma 3.5 need not be V^+. It is only an auxiliary integral model used to build the tensor functor in Section 5.

Example 3.6. The higher BCH terms really occur. In strictly upper triangular 4×4 matrices let $X = E_{12} + E_{23}$ and $Y = E_{34}$. Then $[X,Y] = E_{24}$ and $[X,[X,Y]] = E_{14} \neq 0$, while $[Y,[X,Y]] = 0$, so

$$p^{-2}\,\mathrm{BCH}(p^2X, p^2Y) = X + Y + \frac{p^2}{2}E_{24} + \frac{p^4}{12}E_{14}. \tag{3.10}$$

The coefficient $1/12$ of the cubic term involves exactly the primes 2 and 3, and after scaling by p^2 the term is still integral there, as Lemma 3.1 predicts. Note also that an F-stable subalgebra can be abelian and still be non-normal and have nonzero matrix powers. Abelianness of a quotient does not allow one to truncate the exponential after the linear term.

4. Dieudonné theory and prismatic input

We now collect the p-adic inputs. The results of Section 3 do not depend on them.

4.1. **Perfect fields.** Over a perfect field, displays identify connected p-divisible groups with finite free W-modules L satisfying $L \subseteq \varphi L \subseteq p^{-1}L$, with integral operators φ^{-1} and $p\varphi$ [Zin02]. Saturated F-stable sublattices and their quotients satisfy the same inequalities. In particular each $\mathfrak{l}^+ = \mathfrak{l} \cap \mathfrak{a}^+$, and each quotient lattice in Lemma 3.4, is a completely slope-divisible Dieudonné lattice.

We also need a descent to finite fields. After passing to an algebraic closure of k, each slope factor has a model over a finite field [OZ02]. Choose s so that on a lattice of slope μ the operator

$$U_\mu = p^{-s\mu}\varphi^s$$

is an integral semilinear automorphism. In a U_μ-fixed basis, the matrices of $p\varphi$ and φ^{-1} have entries in $W(\mathbf{F}_{p^s})$, which descends the integral diagram. The normalised φ, which may have denominators, is not used here.

4.2. **Period coordinates.** Let A be a quasi-regular semiperfect $\mathbf{F}_p$-algebra and X a connected p-divisible group. The Dieudonné comparison of D'Addezio and van Hoften [DvH] gives

$$T_pX(A) = \bigl(A_{\mathrm{cris}}(A) \otimes_W \mathbb{D}(X)\bigr)^{\varphi=1}, \tag{4.1}$$

$$\widetilde{X}(A) = \bigl(A_{\mathrm{cris}}(A)[1/p] \otimes_{K_0} \mathbb{D}(X)[1/p]\bigr)^{\varphi=1}. \tag{4.2}$$

These identifications are compatible with composition of internal Homs. Iterating the bilinear comparison handles every bracketed product, and hence, by taking differences, every nested commutator.

Lemma 4.1. *Let $L = p^2\mathfrak{l}^+$ and $N = \Pi(L)$. On the universal qrsp chart let $n = \exp(x)$ be the universal Tate automorphism and*

$$\mathrm{per}(x) \in \bigl(A_{\mathrm{cris}}(A) \otimes_W L\bigr)^{\varphi=1}$$

its period coordinate. Then in the ambient rational endomorphism algebra

$$\mathbb{D}(n) = \sum_{m=0}^{r-1} \frac{\mathrm{per}(x)^m}{m!}, \qquad \log \mathbb{D}(n) = \mathrm{per}(x). \tag{4.3}$$

If $\mathfrak{j} \lhd \mathfrak{l}$ and $\mathfrak{d} = \mathfrak{l}/\mathfrak{j}$ is abelian, with quotient map q, then

$$q\bigl(\log \mathbb{D}(n)\bigr) = \mathrm{per}(q(x)). \tag{4.4}$$

Proof. Compatibility of (4.1) and (4.2) with composition computes the period of each matrix product, hence of each term of the finite exponential, and applying the matrix logarithm gives (4.3). Only after the full logarithm has been formed do we apply q. Since (4.3) lies in $\mathfrak{l}$, the projection makes sense, and functoriality of (4.1) gives (4.4). We never project an individual power of $\mathbb{D}(n) - 1$ to $\mathfrak{d}$. □

The order of operations in this proof is what makes all higher BCH terms cancel before one passes to the abelian quotient.

4.3. **Quasi-syntomicity and algebraisation.** Let $R = k[[t_1, \dots, t_d]]$. Then $\operatorname{Spec} R$ is irreducible, normal, Frobenius-smooth and quasi-syntomic, since absolute Frobenius has the finite syntomic presentation

$$F_*R = R[X_1, \dots, X_d]/(X_1^p - t_1, \dots, X_d^p - t_d). \tag{4.5}$$

If $P \to \operatorname{Spec} R$ is an fpqc torsor under a qrsp group, then P and all its Čech products are quasi-syntomic. Indeed, after the torsor cover the relative cotangent complex is pulled back from that of the group, and Tor-amplitude in $[-1, 0]$ descends along faithfully flat maps.

We use two algebraisation statements:

(1) compatible systems of torsors, and equivariant maps between them, under countable inverse limits of finite group schemes algebraise uniquely over $\operatorname{Spec} R$;
(2) compatible systems of p-divisible groups over the infinitesimal thickenings of the closed point of $\operatorname{Spf} R$ algebraise uniquely, and their Dieudonné crystals on the formal and algebraic bases agree [dJ95].

This lets us apply prismatization to honest quasi-syntomic schemes rather than to formal schemes.

4.4. **The prismatic input.** Two comparison results are used.

(P1) *Prismatization.* For a quasi-syntomic scheme S of characteristic p, vector bundles on the Cartier–Witt stack $S^{\mathbb{\Delta}}$ are identified, symmetric monoidally, with vector bundle crystals on the absolute prismatic site [BL22]. Prismatization takes the qrsp torsors used here to torsors under formal groups.
(P2) *Prismatic and crystalline crystals.* On qrsp charts of characteristic p one has $\mathbb{\Delta}_A = A_{\mathrm{cris}}(A)$, with prismatic ideal generated by p [BS22]. Prismatic vector bundle crystals are thereby identified with crystalline ones, compatibly with Frobenius, tensor operations, descent and horizontal maps.

These are used to show that the torsor built from the BCH period coordinate gives the same isocrystal as the universal p-divisible group (Section 5), and later to identify the isocrystal of the signed quotient representation with that of a Kummer extension (Section 6). The subnormal chain, the integrality of the BCH law, the Frobenius lifting and the final denominator argument do not use them. In particular the prismatic input does not by itself prove the converse. The converse comes from the Kummer construction of Section 6.

4.5. **Finite flat descent of monodromy.** Let $S' \to S$ be a finite flat surjection of irreducible Noetherian Frobenius-smooth schemes and suppose the monodromy in question is unipotent. Then the monodromy group does not change under pullback to S'. We use this to pass from the naive cover $D_{\mathfrak{l},2}$ to $D_{\mathfrak{l}}$.

5. The upper bound

In this section we prove the implication $\Rightarrow$ of (2.7).

Proposition 5.1. *Let $S = \operatorname{Spec} R$ with $R = k[[t_1, \dots, t_d]]$, and let $i\colon S \to D^{\mathrm{alg}}_{\mathfrak{l},2}$ be the algebraisation of a based formal map. There is a based exact symmetric monoidal functor*

$$\Psi_i\colon \operatorname{Rep}_{K_0}(U(\mathfrak{l})) \to \operatorname{Isoc}(S) \tag{5.1}$$

taking the defining representation V to the pullback M_S of the universal isocrystal. Consequently

$$G(M_S, z) \subseteq U(\mathfrak{l}). \tag{5.2}$$

Proof. Pull back the $N_{\mathfrak{l}}$-torsor $T_{\mathfrak{l}} \to D_{\mathfrak{l},2}$, and let $P_i \to S$ be the algebraised torsor. Its universal framing is

$$h_u\colon X_{P_i} \xrightarrow{\sim} Y_{P_i}, \qquad h_{nu} = n \circ h_u. \tag{5.3}$$

Since $N_{\mathfrak{l}}$ is qrsp and all bases and Čech products are quasi-syntomic, prismatization gives an $N_{\mathfrak{l}}^{\mathbb{\Delta}}$-torsor $P_i^{\mathbb{\Delta}} \to S^{\mathbb{\Delta}}$. By Lemma 4.1, the integral period coordinate defines a group homomorphism

$$N_{\mathfrak{l}}^{\mathbb{\Delta}} \to \widehat{U}_{L_{\mathfrak{l}}}. \tag{5.4}$$

This homomorphism is defined before any representation is chosen, the group law on the target being the full integral BCH law. In the defining representation it is

$$n = \exp(x) \mapsto \mathbb{D}(n) = \exp(\operatorname{per}(x)). \tag{5.5}$$

Pushing $P_i^{\mathbb{\Delta}}$ forward along (5.4) gives a $U_{L_{\mathfrak{l}}}$-torsor. For a rational representation of $U(\mathfrak{l})$, choose a stable lattice as in Lemma 3.5 and form the associated vector bundle. After inverting p, the descent matrix for the defining representation is (5.5). Dieudonné theory applied to the framing (5.3) produces the same rational matrix. The auxiliary lattice and V^+ are commensurable, so the two descent data agree after inverting p.

By (P1) and (P2) of Section 4.4, the resulting object is the rational crystalline isocrystal M_S, with its Frobenius and connection. Doing the same for every representation gives (5.1). Exactness and compatibility with tensor products can be checked after the flat torsor cover, where they are statements about vector spaces. The defining representation is faithful and tensor-generates $\operatorname{Rep}(U(\mathfrak{l}))$, so Tannaka duality [DM82] gives the closed immersion (5.2). □

Corollary 5.2. *For every F-stable $\mathfrak{l}$,*

$$G(M_{D_{\mathfrak{l}}}, z) \subseteq U(\mathfrak{l}), \qquad \operatorname{Lie} G(M_{D_{\mathfrak{l}}}, z) \subseteq \mathfrak{l}. \tag{5.6}$$

If $Z \subseteq D_{\mathfrak{l}}$ is based and formally smooth, then $\operatorname{Lie} G(M_Z, z) \subseteq \mathfrak{l}$.

Proof. Apply Proposition 5.1 to $D_{\mathfrak{l},2}$. By Proposition 2.5, the map $D_{\mathfrak{l},2} \to D_{\mathfrak{l}}$ algebraises to a finite flat surjection of regular Frobenius-smooth schemes. The slope-graded pieces of the universal isocrystal are constant, so the monodromy is unipotent, and finite flat descent of monodromy identifies the two monodromy groups. This gives (5.6). The last assertion follows by restriction to Z. □

This proves

$$Z \subseteq D_{\mathfrak{b}} \implies \operatorname{Lie} G(M_Z, z) \subseteq \mathfrak{b}. \tag{5.7}$$

The rest of the paper is devoted to the converse.

6. A Kummer extension for one abelian step

Throughout this section $\mathfrak{j} \lhd \mathfrak{l} \subseteq \mathfrak{a}$ are F-stable and $\mathfrak{d} = \mathfrak{l}/\mathfrak{j}$ is abelian, with quotient map q. Put

$$L^+ = p^2\mathfrak{l}^+, \qquad J^+ = p^2\mathfrak{j}^+, \qquad D^+ = q(L^+) \subseteq \mathfrak{d}. \tag{6.1}$$

Since $\mathfrak{j}^+$ is saturated in $\mathfrak{l}^+$, the lattice $D^+ = L^+/J^+$ is finite free. It is completely slope-divisible with strictly negative slopes. Let

$$Q_0 = X(D^+) \tag{6.2}$$

be the associated connected p-divisible group.

6.1. **The quotient coordinate.** There are exact sequences of fpqc sheaves

$$1 \to \Pi(J^+) \to \Pi(L^+) \xrightarrow{\chi_0} T_pQ_0 \to 0, \tag{6.3}$$

$$1 \to \widetilde{\Pi}(\mathfrak{j}) \to \widetilde{\Pi}(\mathfrak{l}) \xrightarrow{\tilde{q}} \widetilde{Q}_0 \to 0. \tag{6.4}$$

Indeed, the Dieudonné sequence $0 \to X(J^+) \to X(L^+) \to Q_0 \to 0$ is exact on each finite flat level of p^m-torsion, and compatible lifts can be chosen recursively, which gives fpqc surjectivity on Tate modules. Since $\mathfrak{d}$ is abelian, all nonlinear BCH terms die under q, so χ_0 is a homomorphism.

Passing to homogeneous spaces gives a based morphism

$$q_0 \colon Z(L^+) \to Q_0 \tag{6.5}$$

whose scheme-theoretic zero fibre is

$$q_0^{-1}(0) = Z(J^+). \tag{6.6}$$

To see this, lift a point of the fibre to $g \in \widetilde{\Pi}(\mathfrak{l})$. Then $\tilde{q}(g) \in T_pQ_0$, and by (6.3) it lifts to some $n \in \Pi(L^+)$. Now $n^{-1}g \in \widetilde{\Pi}(\mathfrak{j})$, with ambiguity exactly $\Pi(J^+)$. So on every fpqc test ring the fibre is the expected homogeneous space.

6.2. **Lifting through the finite quotient.** The space $D_{\mathfrak{l}}$ is covered by $Z(L^+)$ through a finite connected group scheme. Pulling this cover back along a map from a power series ring can produce a singular scheme, on which the monodromy formalism is unavailable. Instead we kill the finite connected group by a power of Frobenius.

Lemma 6.1. *Let H be a finite connected group scheme over k and $\pi \colon \operatorname{Spf} B_2 \to \operatorname{Spf} B$ a finite faithfully flat H-torsor of complete local rings with residue field k. Let $i \colon \operatorname{Spf} R \to \operatorname{Spf} B$ be based, with $R = k[[t_1, \dots, t_d]]$. Then for some $s \geq 0$ the relative Frobenius*

$$f \colon \operatorname{Spf} R^{(-s)} \to \operatorname{Spf} R, \qquad f^{\#}(a) = a^{p^s} \otimes 1, \tag{6.7}$$

lifts to a based map $\ell \colon \operatorname{Spf} R^{(-s)} \to \operatorname{Spf} B_2$ with $\pi\ell = if$. The map f is finite and faithfully flat, and $R^{(-s)}$ is again a power series ring.

Proof. Let I be the augmentation ideal of $\mathcal{O}(H)$ and choose s with $a^{p^s} = 0$ for all $a \in I$. If $c \colon B_2 \to \mathcal{O}(H) \otimes B_2$ is the coaction, then $c(b) - 1 \otimes b \in I \otimes B_2$, so $c(b^{p^s}) = 1 \otimes b^{p^s}$. Thus b^{p^s} is invariant and lies in B. Define

$$\ell^{\#}(b) = i^{\#}(b^{p^s}) \otimes 1. \tag{6.8}$$

The Frobenius twist makes $\ell^{\#}$ k-linear, and $\pi\ell = if$ holds on B. The relative Frobenius of a power series ring is finite free, with basis the monomials with exponents $< p^s$. □

Since f is based, pullback along f induces a closed immersion of Tannakian groups, so a containment of Lie algebras in $\mathfrak{j}$ is preserved.

6.3. **The signed quotient representation.** Let $U(\mathfrak{l})$ act on $W_{\mathfrak{d}} = K_0 e \oplus \mathfrak{d}$ by

$$\rho_{\mathfrak{d}}^{-}(\exp x)\, e = e - q(x), \qquad \rho_{\mathfrak{d}}^{-}(\exp x)\, v = v \quad (v \in \mathfrak{d}). \tag{6.9}$$

This is a representation because $\mathfrak{d}$ is abelian. Give the fibre the Frobenius $\Phi(e) = e$, $\Phi|_{\mathfrak{d}} = \varphi_{\mathfrak{d}}$. The kernel of $\rho_{\mathfrak{d}}^{-}$ is $U(\mathfrak{j})$, and the integral BCH group preserves $We \oplus D^{+}$.

Now let $u\colon S = \operatorname{Spf} R \to Z(L^{+})$ be based and put $q_S = q_0 u\colon S \to Q_0$. For the Artinian quotients $R_n = R/\mathfrak{m}^{n+1}$ define the Kummer extension

$$E_n = (Q_{0,R_n} \times \mathbf{Z}[1/p])/\mathbf{Z}, \qquad \ell \mapsto ([\ell]q_n, \ell). \tag{6.10}$$

Its p^m-torsion over j/p^m is the finite flat torsor of roots

$$[p^m]v = [j]q_n. \tag{6.11}$$

The finite levels and the transition maps $[p]$ glue to a formal p-divisible group, which algebraises uniquely; call it E_{q_S}.

Proposition 6.2. *The object of $\langle M_S \rangle$ associated with $\rho_{\mathfrak{d}}^{-}$ is canonically isomorphic, compatibly with connection and Frobenius, to $\mathbb{D}(E_{q_S})[1/p]$.*

Proof. Pull back the $\Pi(L^{+})$-torsor P from $Z(L^{+})$ and push it forward along $\chi_0\colon \Pi(L^{+}) \to T_p Q_0$. The result is canonically the torsor of roots of q_S; on representatives the isomorphism is

$$[a, g] \mapsto a + \tilde{q}(g), \tag{6.12}$$

which is equivariant and becomes a translation after any fpqc trivialisation.

A system of roots $u = (u_m)$ splits E_{q_S}. If h_u is the corresponding covariant framing and $t \cdot u = (u_m + t_m)$ is translation by t, then

$$h_{t\cdot u} = A_{-t} h_u, \qquad A_t(v, c) = (v + t(c), c), \tag{6.13}$$

so the covariant Dieudonné cocycle of the frames is $1 - \mathbb{D}(t)$. On the source torsor write $n = \exp(x)$. By Lemma 4.1,

$$\log \mathbb{D}(n) = \mathbb{D}(x), \qquad q(\log \mathbb{D}(n)) = \mathbb{D}(\chi_0(n))(e). \tag{6.14}$$

Applying the signed representation after the full logarithm gives

$$\rho_{\mathfrak{d}}^{-}(\mathbb{D}(n)) = 1 - \mathbb{D}(\chi_0(n)). \tag{6.15}$$

The operators on the target square to zero, so (6.15) is the whole Čech cocycle of (6.13). The higher terms on the source have already been absorbed in $\log \exp(x) = x$ before q is applied.

It remains to algebraise the torsors and apply (P1) and (P2). On the common prismatic torsor both objects are the same constant rational module with the same descent matrix (6.15). Compatibility with connection and Frobenius holds on the cover, and therefore descends. □

6.4. **Splitting forces the coordinate to vanish.**

Lemma 6.3. *Let $Q = X(D^{+})$ be connected, completely slope-divisible, of strictly negative slopes, and let $q\colon \operatorname{Spf} R \to Q$ be based, with $R = k[[t_1, \dots, t_d]]$. If the F-isocrystal $\mathbb{D}(E_q)[1/p]$ on $\operatorname{Spec} R$ splits, then $q = 0$.*

Proof. The rational Dieudonné functor is fully faithful over the normal Frobenius-smooth base $\operatorname{Spec} R$. A splitting of isocrystals therefore comes from a quasi-homomorphism

$$s \in \operatorname{Hom}(C_{\mathrm{et}}, E_q) \otimes_{\mathbf{Z}_p} \mathbf{Q}_p, \qquad C_{\mathrm{et}} = \mathbf{Q}_p/\mathbf{Z}_p,$$

with $\pi_E s = 1$. Since s is a single element of a single Hom group, one power of p clears its denominator:

$$s_N = p^N s \in \operatorname{Hom}(C_{\mathrm{et}}, E_q), \qquad \pi_E s_N = [p^N]. \tag{6.16}$$

This equality holds integrally because Hom groups of p-divisible groups are p-torsion-free. By (6.16) the Kummer extension of $r = [p^N]q$ splits, and so r has a compatible system of p^m-th roots for all m. The universal cover of Q is its perfection Q^{perf}. So if $x_1, \dots, x_e$ are parameters on Q, then for every m

$$r^{\#}(x_i) \in R^{p^m}. \tag{6.17}$$

But

$$R^{p^m} = k[[t_1^{p^m}, \dots, t_d^{p^m}]], \qquad \bigcap_{m \geq 0} R^{p^m} = k. \tag{6.18}$$

As r is based, $r^{\#}(x_i)$ also lies in the maximal ideal, so it is zero and $[p^N]q = 0$. Hence q factors through the finite connected group scheme $Q[p^N]$. Its augmentation ideal is nilpotent and R is reduced, so the based map q factors through the identity section, i.e. $q = 0$. □

6.5. **The converse for one step.**

Proposition 6.4. *Let $\mathfrak{j} \lhd \mathfrak{l}$ with abelian quotient, and let $i\colon Z = \operatorname{Spf} R \to D_{\mathfrak{l}}$ be based, with $R = k[[t_1, \dots, t_d]]$. If*

$$\operatorname{Lie} G(M_Z, z) \subseteq \mathfrak{j}, \tag{6.19}$$

then i factors through $D_{\mathfrak{j}}$.

Proof. Apply Lemma 6.1 to the finite connected torsor $Z(p^2\mathfrak{l}^+) \to D_{\mathfrak{l}}$. This gives a finite faithfully flat relative Frobenius $f\colon Z^{(-s)} \to Z$ and a lift $\ell\colon Z^{(-s)} \to Z(L^+)$ of if, and (6.19) still holds on $Z^{(-s)}$. Put $q' = q_0\ell\colon Z^{(-s)} \to Q_0$.

By Proposition 6.2, the Kummer object $\mathbb{D}(E_{q'})[1/p]$ lies in the tensor category generated by $M_{Z^{(-s)}}$ and corresponds to a representation with kernel $U(\mathfrak{j})$. Its monodromy group is therefore an algebraic subgroup of the vector group $\mathfrak{d}$ with zero Lie algebra. Such a subgroup is finite, and a vector group in characteristic zero has no nontrivial finite subgroups. So the Kummer isocrystal has trivial monodromy, and the splitting of its fibre at the base point extends to a horizontal splitting. The defect of Frobenius-equivariance is horizontal and vanishes on the fibre, so the splitting is Frobenius-equivariant.

By Lemma 6.3, $q' = 0$, and by (6.6) ℓ factors through $Z(J^+)$. Hence if lands in the closed subspace $D_{\mathfrak{j}}$. If I is the ideal of $D_{\mathfrak{j}}$, then $f^{\#}i^{\#}(I) = 0$. The ring map $f^{\#}$ is finite and faithfully flat, hence injective, so $i^{\#}(I) = 0$. Thus i itself factors through $D_{\mathfrak{j}}$. □

7. Proof of the main theorem

7.1. **The converse.**

Let $\mathfrak{b} \subseteq \mathfrak{a}$ be arbitrary and take the chain of Lemma 3.4,

$$\mathfrak{a} = \mathfrak{h}_1 \supseteq \mathfrak{h}_2 \supseteq \cdots \supseteq \mathfrak{h}_r = \mathfrak{b}. \tag{7.1}$$

Proposition 7.1. *For every based formally smooth closed $Z \subseteq \operatorname{Def}^{\mathrm{sus}}(Y)$,*

$$\operatorname{Lie} G(M_Z, z) \subseteq \mathfrak{b} \implies Z \subseteq D_{\mathfrak{b}}. \tag{7.2}$$

Proof. We have $D_{\mathfrak{h}_1} = D_{\mathfrak{a}} = \mathrm{Def}^{\mathrm{sus}}(Y)$. Suppose $Z \subseteq D_{\mathfrak{h}_i}$. Then $\mathrm{Lie}\, G(M_Z, z) \subseteq \mathfrak{b} \subseteq \mathfrak{h}_{i+1}$, and by Lemma 3.4 $\mathfrak{h}_{i+1} \lhd \mathfrak{h}_i$ with abelian quotient whose lattice is completely slope-divisible of negative slope. So Proposition 6.4 gives $Z \subseteq D_{\mathfrak{h}_{i+1}}$. After at most $r-1$ steps we reach $D_{\mathfrak{b}}$. At each step the vanishing of the defining ideal is descended through an injective Frobenius map; we never argue only after perfection. □

Corollary 5.2 and Proposition 7.1 together give (2.7).

7.2. **Exact monodromy.** It remains to prove (2.8). Let $\mathfrak{c} = \mathrm{Lie}\, G(M_{D_{\mathfrak{b}}}, z)$. It is F-stable, and $\mathfrak{c} \subseteq \mathfrak{b}$ by Corollary 5.2. Applying Proposition 7.1 to $D_{\mathfrak{b}}$ itself gives $D_{\mathfrak{b}} \subseteq D_{\mathfrak{c}}$, and $\mathfrak{c} \subseteq \mathfrak{b}$ gives a closed immersion $D_{\mathfrak{c}} \hookrightarrow D_{\mathfrak{b}}$. So the two formal schemes coincide. We have to show that this forces $\mathfrak{c} = \mathfrak{b}$.

Write $L_{\mathfrak{l}} = p^2\mathfrak{l}^+$ and $P = L_{\mathfrak{b}}/L_{\mathfrak{c}}$. It is finite free with negative slopes. For every $\mathfrak{l}$,

$$\dim D_{\mathfrak{l}} = \dim X(L_{\mathfrak{l}}) = \mathrm{length}_W(L_{\mathfrak{l}}/\varphi^{-1}L_{\mathfrak{l}}), \tag{7.3}$$

and the snake lemma applied to the saturated exact sequence $0 \to L_{\mathfrak{c}} \to L_{\mathfrak{b}} \to P \to 0$ gives

$$\dim D_{\mathfrak{b}} - \dim D_{\mathfrak{c}} = \mathrm{length}_W(P/\varphi^{-1}P). \tag{7.4}$$

If $P \neq 0$, then φ^{-1} is topologically nilpotent on P because the slopes are negative. If $P = \varphi^{-1}P$, iterating would give $P \subseteq pP$, which contradicts Nakayama's lemma. So the right side of (7.4) is positive. Since $D_{\mathfrak{b}} = D_{\mathfrak{c}}$, it is zero, and therefore $P = 0$ and $\mathfrak{c} = \mathfrak{b}$. This completes the proof of Theorem 2.3.

The degenerate cases are covered as well. If $\mathfrak{b} = 0$ then $D_0 = \mathrm{Spf}\, k$ and the monodromy is trivial. If $r = 1$ then $\mathfrak{a} = 0$ and $\mathrm{Def}^{\mathrm{sus}}(Y) = \mathrm{Spf}\, k$.

8. The questions of D'Addezio and van Hoften

8.1. **Conjecture 5.5.7.** For completely slope-divisible Y, D'Addezio and van Hoften conjecture that for every F-stable Lie subalgebra $\mathfrak{b} \subseteq \mathfrak{a}$

$$Z \subseteq Z(\mathfrak{b}^+) \iff \mathrm{Lie}\, G(M_Z) \subseteq \mathfrak{b}, \tag{8.1}$$

and that

$$\mathrm{Lie}\, G(M_{Z(\mathfrak{b}^+)}) = \mathfrak{b}. \tag{8.2}$$

They prove the implication from containment of spaces to containment of monodromy. Theorem 2.3 supplies the converse and the exact monodromy in the formally smooth case, with the strongly Tate-linear space taken to be $D_{\mathfrak{b}} = Z(p^2\mathfrak{b}^+)/H_{\mathfrak{b},2}$. We record this as follows.

Theorem 8.1. *Let k be perfect, p any prime, and Y/k completely slope-divisible with any finite number of slopes. For every F-stable Lie subalgebra $\mathfrak{b} \subseteq \mathfrak{a}$ and every based formally smooth closed formal subscheme $Z \subseteq \mathrm{Def}^{\mathrm{sus}}(Y)$, the equivalence (8.1) holds with $D_{\mathfrak{b}}$ in place of $Z(\mathfrak{b}^+)$, and (8.2) holds for $D_{\mathfrak{b}}$.*

We do not treat singular or nonreduced Z. These are not needed for Question 8.7.1.

8.2. **Question 8.7.1.** Let $C = C(Y, \lambda)$ be a central leaf in a Siegel modular variety and $Z \subseteq C$ an irreducible smooth closed subvariety. D'Addezio and van Hoften observe that their local monodromy theorem makes the local monodromy group independent of the point, so that Conjecture 5.5.7 would answer Question 8.7.1. We give the deduction in detail, including the dimension count.

Theorem 8.2. *In the setting of Theorem 8.1, Question 8.7.1 has an affirmative answer: if Z is strongly Tate-linear at one closed point, it is strongly Tate-linear at every closed point.*

Proof. Suppose Z is strongly Tate-linear at z_0. With the standard framing of the sustained deformation space at z_0, there is an F-stable Lie subalgebra $\mathfrak{b}_{z_0}$ with

$$\widehat{Z}_{z_0} = D_{\mathfrak{b}_{z_0}}. \tag{8.3}$$

By Theorem 8.1,

$$\operatorname{Lie} G(M_{\widehat{Z}_{z_0}}, z_0) = \mathfrak{b}_{z_0}. \tag{8.4}$$

By the local monodromy theorem of [DvH], the local monodromy at any closed point $z \in Z$ is the same unipotent radical of the global monodromy group. Transporting the fibre along an isomorphism of p-divisible groups in the central leaf, let $\mathfrak{b}_z$ be the corresponding F-stable Lie algebra. It is the transport of (8.4). In particular the negative-slope Dieudonné data attached to it, and hence $\dim D_{\mathfrak{b}_z}$, do not depend on z.

The converse direction of Theorem 8.1, applied to the power series formal scheme $\widehat{Z}_z$, gives

$$\widehat{Z}_z \subseteq D_{\mathfrak{b}_z}. \tag{8.5}$$

As Z is smooth and irreducible,

$$\dim \widehat{Z}_z = \dim Z = \dim \widehat{Z}_{z_0} = \dim D_{\mathfrak{b}_{z_0}} = \dim D_{\mathfrak{b}_z}. \tag{8.6}$$

Both sides of (8.5) are formal power series schemes. A nonzero ideal in a regular local domain has positive height, so a closed immersion of regular local formal schemes of the same dimension is an isomorphism. Hence $\widehat{Z}_z = D_{\mathfrak{b}_z}$, and Z is strongly Tate-linear at z. □

8.3. **Question 8.7.2: reduction to a realisation problem.** Question 8.7.2 asks whether a smooth irreducible Z, strongly Tate-linear at one point, is an irreducible component of a central leaf in the reduction of a Hodge-type Shimura variety. The general case remains open, and the difficulty is not one of local monodromy. By the results above, the completed local rings of Z are all described by one F-stable Lie subalgebra $\mathfrak{b}$. The question is whether this Lie algebra comes from global Hodge tensors. Chai [Cha24] asks the corresponding local question: which Tate-linear formal subvarieties of a Siegel central leaf are formal germs of intersections with Shimura subvarieties?

Proposition 8.3. *Let $Z \subseteq C$ be irreducible, smooth, closed, and strongly Tate-linear at z_0, with $\widehat{Z}_{z_0} = D_{\mathfrak{b}}$. Suppose there are a Hodge-type Shimura subvariety $S \hookrightarrow \mathscr{A}_g$ with good reduction at p and an irreducible component C_S^0 of a central leaf of $S_{\mathbf{F}_p}$ such that, as closed formal subschemes of $\widehat{C}_{z_0}$,*

$$\widehat{(C_S^0)}_{z_0} = D_{\mathfrak{b}}. \tag{8.7}$$

Then $Z = C_S^0$ near z_0, and if both are closed irreducible subvarieties of C, then $Z = C_S^0$. In particular Question 8.7.2 has an affirmative answer for Z.

Proof. By (8.7) the ideals of Z and C_S^0 in $\mathcal{O}_{C,z_0}$ become equal after completion. Completion of a Noetherian local ring is faithfully flat, so they are already equal in $\mathcal{O}_{C,z_0}$, and the two subvarieties agree on a neighbourhood of z_0. Two closed irreducible subvarieties of C with a common nonempty open subset have the same closure, so they are equal. □

Thus Question 8.7.2 reduces to realising $\mathfrak{b}$ globally by Hodge tensors. We now give cases where this can be done.

8.4. **Maximal local monodromy.**

Theorem 8.4. *Let $Z \subseteq C$ be as above, and suppose that at one point, equivalently by Theorem 8.2 at every point, the Lie algebra of Z is $\mathfrak{b} = \mathfrak{a}$. Then Z is the irreducible component of C through that point. In particular Question 8.7.2 has an affirmative answer.*

Proof. For $\mathfrak{b} = \mathfrak{a}$ we have $D_{\mathfrak{a}} = \mathrm{Def}^{\mathrm{sus}}(Y) = \widehat{C}_{z_0}$, so $\widehat{Z}_{z_0} = \widehat{C}_{z_0}$. By faithful flatness of completion, the ideals agree in $\mathcal{O}_{C,z_0}$, so Z contains an open neighbourhood of z_0 in C. A central leaf is smooth, so its irreducible components are open and closed. As Z is closed and irreducible, it is the component through z_0. For the Hodge-type Shimura variety one takes the Siegel modular variety itself. □

By Theorem 2.3, the hypothesis $\mathfrak{b} = \mathfrak{a}$ is equivalent to $\operatorname{Lie} G(M_Z, z_0) = \mathfrak{a}$.

8.5. **The ordinary case in products of GSpin Shimura varieties.** On the ordinary locus $\mathfrak{a}$ is abelian, and strong Tate-linearity is the Serre–Tate condition that the completed germ is a formal subtorus. Jiang [Jia23] proved Chai's Tate-linear conjecture, and the stronger characteristic-p Ax–Schanuel statement, for subvarieties whose map to $\mathscr{A}_g$ factors through the ordinary locus of a product of GSpin Shimura varieties.

Theorem 8.5. *Let p be odd and let $C = \mathscr{A}_g^{\mathrm{ord}}$, with a suitable prime-to-p level. Let $Z \subseteq C$ be an irreducible smooth closed subvariety, strongly Tate-linear at one point, and suppose that $Z \hookrightarrow \mathscr{A}_g^{\mathrm{ord}}$ factors through $\mathscr{S}_{I,\mathbf{F}_p}^{\mathrm{ord}}$, where $\mathscr{S}_I$ is the good integral model, under a Hodge embedding, of a product of* GSpin *Shimura varieties with nonempty ordinary locus. Then Z is an irreducible component of a central leaf in the reduction of a Hodge-type Shimura variety.*

Proof. The ordinary Serre–Tate formal group is commutative, so the strongly Tate-linear germ of Z is a formal subtorus and Z is Tate-linear in Chai's sense. By Jiang's theorem for products of GSpin Shimura varieties, Z is special: its Zariski closure is an irreducible component of the reduction of a Hodge-type special subvariety $S' \subseteq \mathscr{A}_g$ in characteristic zero.

Since Z lies in the ordinary locus, it is an irreducible component of the ordinary part of the relevant component of the reduction of S'. The ordinary part of a Hodge-type Shimura variety, when nonempty for the given Hodge embedding, is its μ-ordinary locus [Wor13]. The μ-ordinary locus is a central leaf: the Dieudonné module with its Hodge tensors is constant up to isomorphism along it [Zha17]. Hence the ordinary component containing Z is a central leaf of $S'_{\mathbf{F}_p}$, and Z is one of its irreducible components. □

Corollary 8.6. *Let p be odd. Every irreducible smooth closed subvariety of $\mathscr{A}_2^{\mathrm{ord}}$ which is strongly Tate-linear at one point satisfies the conclusion of Question 8.7.2.*

Proof. Through the exceptional isomorphism between split $\mathrm{GSpin}(3,2)$ and GSp_4, the Siegel threefold is itself a GSpin Shimura variety, so the factorisation hypothesis of Theorem 8.5 holds automatically. □

More generally, Theorem 8.5 applies to ordinary subvarieties of products of GSpin Shimura varieties, including the standard examples covered by Jiang's theorem.

8.6. **A liftable ordinary case.** There is an older case of a different kind. Moonen [Moo98] proved that an irreducible algebraic subvariety in characteristic zero of a Siegel modular variety with good integral model is a Shimura subvariety of Hodge type if it is formally linear at an ordinary point of the special fibre.

Proposition 8.7. *Let $C = \mathscr{A}_g^{\mathrm{ord}}$. Suppose Z is the ordinary special-fibre component of the flat closure of an irreducible subvariety $Z_\eta \subseteq \mathscr{A}_{g,\eta}$, and that the formal germ of this closure at z_0 is formally linear in Moonen's sense. Then Z satisfies the conclusion of Question 8.7.2.*

Proof. By Moonen's theorem Z_η is a Shimura subvariety of Hodge type. Its ordinary special fibre is its μ-ordinary locus, hence a central leaf, and Z is the relevant irreducible component by hypothesis. □

The lifting hypothesis is a strong one. This is why Moonen's theorem gives evidence for Question 8.7.2 but does not settle the question in characteristic p.

8.7. **What remains.** Local monodromy is no longer the obstruction. If Z is strongly Tate-linear at one point, Theorem 8.2 propagates this to every point, and Theorem 2.3 identifies the local Lie algebra with the local monodromy Lie algebra. What is left of Question 8.7.2 is the following:

(8.8) which F-stable negative Lie subalgebras $\mathfrak{b}$ can be realised globally by Hodge tensors?

This is a noncommutative, non-ordinary version of Chai's question of which Tate-linear formal subtori come from Shimura subvarieties. We expect the next step to be a global realisation theorem for $\mathfrak{b}$ by Hodge tensors, rather than further local computations.

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